

CSD, Rennes

TD 3

September 27, 2023

1 Pole placement and state observer

1.1 Make a state-feedback

Following TD2, rewrite it as

$$\frac{Y(s)}{U(s)} = \frac{s+1}{s^3 + 20s^2 + 100s}.$$

Then the canonical controllable form is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -100 & -20 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [1 \quad 1 \quad 0].$$

The system is of order 3, and we have to choose three poles.

Let us approximate it as a second-order system and then use the second pole to compensate the zero. Let the second-order system's poles satisfy $s^2 + 2\xi\omega_0 s + \omega_0^2 = 0$. For info, the roots are

$$-\omega_0 \left(\xi \pm i\sqrt{1 - \xi^2} \right).$$

To have the 5% overshoot, we set $\xi = 0.7$. Then $\omega_0 t_r \approx 3$ and from $t_r = 2$ we obtain $\omega_0 = 1.5$. The last pole is -1 . To get the gains, we compute `K=place(A,B,poles)`.

To get ℓ , we have

$$y_\infty = -C(A - BK)^{-1} B \ell y^r \implies \ell = \frac{-1}{C(A - BK)^{-1} B}.$$

1.2 Is it observable?

We have

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -100 & -19 \end{bmatrix},$$

and it is of full rank. Moreover, any minimum realization is observable.

1.3 State observer

Actually, as it is a transfer function, the states depends on our choice and are not measured. We need to reconstruct them in the chose basis using only u and y .

A state observer is of the form

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

or

$$\dot{\hat{x}} = (A - LC)\hat{x} + Bu + Ly = (A - LC)\hat{x} + \begin{bmatrix} B & L \end{bmatrix} \begin{bmatrix} u \\ y \end{bmatrix}.$$

Defining $e_x := x - \hat{x}$, we have $\dot{e}_x = (A - LC)e_x$. Thus, we want to choose L such that the closed-loop poles are OK.

Usually, we prefer to have the closed-loop poles of an observer faster than the closed-loop poles, e.g., 3 times faster.

Note that the eigenvalues of $A - LC$ and $A^\top - C^\top L^\top$ are the same. Thus, we can use the `place` command as `L = (place(A',C',observer_poles))'`. If some poles are the same, `place` will not work, and a minor adjustment is required.

2 LQ and state-observer

2.1 LQ design

To get the LQ control, we need to specify Q and R in the cost function

$$\int_t^\infty (x^\top Qx + u^\top Ru) d\tau.$$

We obviously have $R = \alpha = 10^{-4}$. What about C ? Note that

$$y^2 = (Cx)^2 = x^\top C^\top Cx \implies Q = C^\top C = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We have previously checked that A, B and A, C are controllable and observable, respectively. So, we compute `K2=lqr(A,B,Q,R)`.

2.2 New observer

The poles of the LQ control are the eigenvalues of $(A - BK_2)$. To have our observer consistent, we choose the desired poles 3 times faster and recompute L .

2.3 Simulations

Simulations

3 Noise and state observer

3.1 State-space

The output and the input are noised. Then

$$\dot{x} = Ax + B(u + v_u) = Ax + Bu + v,$$

where $v = Bv_u$. The variance of v_u is du_0 , and the variance of v is $E(vv^\top) = BB^\top du_0 = Q_0$.

The output is

$$y = Cx + w,$$

where the variance of w is dy_0 ; there is no weighting here.

3.2 Conditions

For the existence, we need that A, C is detectable – it is satisfied since A, C is observable. The second condition is that A, J is controllable, where $JJ^\top = Q_0$. In our case, $J = B\sqrt{du_0}$, and the condition is satisfied for $du_0 > 0$.

3.3 Computation

To compute the gain, we use `lqe`. There are three ways.

- The first is `L = lqe(A, B, C, du0, dy0, 0)`, where we explicitly say that the state noise enters with the vector B .
- The second is `L = lqe(A, eye(3), C, Q, dy0)`, where the state noise v acts on the whole state vector and Q is defined as `Q = B*B'*du0`.
- And finally we can use the duality with `LQ` by `L=(lqr(A',C',Qkal,Rkal))'`.

3.4 Simulations

Simulations